

Introductory Astronomy Reading Group

Designed by: Neil Ghugare

Textbook

[Astronomy](#) by Andrew Fraknoi, David Morrison, and Sidney Wolff.

This textbook is 100% free, open-source, college-level, and peer-reviewed. It is written in a conversational tone and has good figures and imagery, making it good for an introductory reading groups. It is more conceptual than math-heavy.

Meetings

Meetings will be held virtually on Discord, **TBA**.

Meetings will be partially recaps of the readings with time for group discussions on topics and points of confusion. Then we will continue our conceptual learning through thoughtful discussions and thought-provoking questions. We will also try to sprinkle in a few calculation-based problems as well, when necessary and interesting. The intention is that the math done during the synchronous meetings will be *no higher* than an introductory calculus level (basic integrals and derivatives). Topics regarding higher levels of mathematics (vector calculus, differential equations, etc.) will be regarded, but the derivational mathematics may be done at a high level.

Meetings will generally follow a 60-minute live session. The meetings in general will follow the format below:

1. *Icebreaker & Astro-News* (10 minutes): I will show and discuss a recent astronomy development or headline pertaining to the topic at hand, and we can discuss.
2. *Conceptual Recap* (15 minutes): We will do a high-level, high-yield summary of the week's major concepts and graphics from the text. The main point of this is to hit key ideas and concepts, not to be a lecture.
3. *Thoughtful Discussion* (15 minutes): I will introduce a few big-picture prompts regarding the topic at hand and we can engage in group discussion or civil debate. Most of these are to help all of us think more deeply, critically, and intimately with the information. Maybe we'll even do some fun games.
4. *The Physics Sprinkle* (15 minutes): Together, we will either solve 1 or 2 quantitative or qualitative problems from the text to bridge the astronomy concepts with core physics concepts, or simply discuss physics relationships.
5. *Looking Ahead* (5 minutes): Wrap up, confirm the next reading assignment, etc.

Content & Calendar

I intend readings to be roughly *50 pg/wk*. This may seem like a lot, but the textbook has a lot of large images and worked examples, which take up more space, inflating the page count. Additionally, most of the reading can be skimmed for core concepts, and does not need to be read super finely. *Please see the notes after the table on the readings too.*

Topic	Details	Reading Sections (OpenStax Astronomy 2e)
Week 1: Our Cosmic Neighborhood	Getting our bearings in the night sky, constellations, the celestial sphere, seasons, lunar phases, and the historical shift from geocentric to heliocentric models.	Chapter 1: All sections (Brief Tour) Chapter 2: 2.1 (Celestial Sphere), 2.2 (Planetary Motion), 2.4 (History of Astronomy) Chapter 4: 4.1 (Earth & Sky), 4.2 (Seasons), 4.5 (Phases & Motions of the Moon) ~65 pg.
Week 2: The Solar System	Overview of solar system formation and comparative planetology. Group members will choose one specific world (e.g., Mars, Europa, Enceladus) to dive into.	Chapter 7: 7.1 (Overview), 7.2 (Composition & Structure) Chapter 14: 14.3 (Formation of the Solar System) Skim Chapters 8–13 and pick one specific planet/moon of interest. ~30 pg (varies).
Week 3: Engines of the Universe	The physics and lifecycle of stars. We will explore hydrostatic equilibrium, Einstein's mass-energy equivalence, how the H-R diagram plots stellar temperature vs. luminosity, and the evolutionary path of a star from the Main Sequence to a Red Giant.	Chapter 15: 15.1 (Structure of the Sun) Chapter 16: 16.2 (Mass, Energy, Relativity) Chapter 18: 18.2 (Measuring Stellar Masses), 18.4 (The H-R Diagram) Chapter 22: 22.1 (Main Sequence to Red Giant)

		Chapter 23: 23.1 (Death of Low-Mass Stars), 23.2 (Evolution of Massive Stars) ~55 pg.
Week 4: Galaxies & Expanding Universe	Structure and dynamics of the Milky Way, classification of external galaxies, the cosmic distance ladder, and Hubble's Law demonstrating the expansion of space.	Chapter 25: 25.1 (Architecture of the Galaxy), 25.3 (Mass of the Galaxy) Chapter 26: 26.1 (Discovery of Galaxies), 26.2 (Types of Galaxies), 26.5 (The Expanding Universe) ~28 pg.
Week 5: The Big Bang & Cosmology	The origin of everything, cosmic microwave background (CMB) radiation, the geometry of spacetime, and the mysterious roles of Dark Matter and Dark Energy.	Chapter 29: 29.1 (The Age of the Universe), 29.2 (A Model of the Universe), 29.3 (The Beginning of the Universe), 29.4 (The Cosmic Microwave Background), 29.5 (What is the Universe Really Made Of?) ~34 pg.
Week 6: " <i>What the heck is going on?</i> "	Modern observational techniques, astronomical instrumentation, active galactic nuclei (AGN), supermassive black holes, and exciting recent breakthroughs in the field.	Chapter 6: All sections (Astronomical Instruments) Chapter 27: 27.1 (Quasars), 27.2 (Supermassive Black Holes) ~48 pg. <i>Additional reading may be assigned.</i>
Additional Weeks (If Applicable)	Any additional weeks on specialized topics: exoplanets, astrobiology, astroparticle, deeper planetary science, black holes, order-of-magnitude estimation, gas and dust/the ISM, etc.	<i>TBA</i>

Notes on specific weeks' readings:

Week 1: Our Cosmic Neighborhood

- **Note: we are skipping Chapter 3 entirely.** Chapter 3 entails Newtonian gravity, Kepler's laws, and basic orbital mechanics. While fundamental, we are not requiring it, but it may be useful to skim if you have no prior history or knowledge with gravitation. An introductory physics textbook could also work as a good supplement.
- **Focus:** Focus heavily on the *geometry* of the sky in Chapter 2 (zenith, ecliptic, etc.) and the visual mechanics of why we see phases and seasons in Chapter 4.
- **Connection to Physics:** Pay attention to how ancient astronomers used basic geometry and trigonometry to estimate sizes and distances long before telescopes existed.

Week 2: The Solar System

- **Note: this is a choose-your-own-adventure week.** Do not try to read all of Chapters 8 through 13 page-by-page. It's a lot of geological data. Pick a few sections that interest you and read only those.
- **Focus:** Everyone will be reading Chapter 7 for the big-picture physics of our solar system's organization. Then pick one specific moon, planet, etc. that fascinates you and prepare a 2-minute "sales pitch" on why yours is the most interesting. We will likely share (voluntarily) and decide whose is the most interesting. You are not required to prepare any visuals for your pitch.
- **Connection to Physics:** In Chapter 14, focus on or investigate the conservation of *angular momentum* (how a giant, slowly rotating cloud of gas collapsed into a rapidly spinning, flat planetary disk).

Week 3: Engines of the Universe

- **Note: we are absolutely skimming the surface of stellar physics.** This textbook has several chapters on stellar classifications and spectroscopy that we are skipping to keep the reading light. Stellar astronomy is a huge field with tons of specializations. Undergraduates typically will take at least a semester of introductory stellar astrophysics, and may even take another full semester of advanced stellar astrophysics. Graduate students also sometimes take multiple semesters of stellar astronomy.
- **Focus:** Focus on the delicate balancing act of *hydrostatic equilibrium* (gravity pushing inward vs. thermal/radiation pressure pushing outward). Look closely at Section 16.2 to see how the proton-proton chain converts mass defect into raw energy via Einstein's equation.
- **Connection to Physics:** Treat the H-R Diagram as a thermodynamic phase diagram for stars. Notice how a star's position on the graph is completely determined by its mass, surface temperature, and luminosity.

Week 4: Galaxies & The Expanding Universe

- **Note: this week is highly theoretical.** This is where astronomy meets particle physics and general relativity.
- **Focus:** Focus deeply on the *Cosmic Microwave Background (CMB)* radiation. It is essentially the “afterglow” of the Big Bang. Understand why it is the ultimate proof of the early, hot, dense universe.
- **Connection to Physics:** Pay attention to the ratios. The text explains that regular matter (everything we can see, touch, and measure) makes up only about 5% of the universe. The rest is Dark Matter (~27%) and Dark Energy (~68%). Think about the implications of the physics we *don't* fully understand yet.

Week 6: “What the heck is going on?”

- **Note: This is our capstone “current events” week.** It bridges historical data with what is actively being published in journals today. Alongside the readings, we will likely discuss why JWST/Rubin are causing some fields of astronomy to “blow up” with unexplainable observations and phenomena, and how astronomers are pushing the boundaries of that knowledge with experiments and supercomputers.
- **Focus:** Look at Chapter 6 to see why we put telescopes in space. Then, look at Chapter 27 to see how supermassive black holes power *Quasars*.
- **Connection to Physics:** This week is all about modern engineering meeting extreme astrophysics. Think about how we use gravitational lensing or adaptive optics to probe the farthest reaches of reality. Also, while not in the text, think about how this relates to overarching problems we as scientists have: writing grant proposals, funding, marketing your experiments, how to navigate the unknown, etc.